

Immanuel Eben Haezer Joseph Aletheia

Software Engineer | Final-Year Electrical Engineering Student

Bandung Regency, West Java 40257, Indonesia • +62 858-2929-0950 • nue.immanuel@gmail.com
linkedin.com/in/immanuel-eben-haezer-joseph-aletheia • github.com/EintsWaveX

SUMMARY

Final-year Electrical Engineering student (GPA 3.77/4.00, 125 of 145 credits completed) working primarily as a software engineer in Python and C/C++, across computer vision, application development, and web-connected systems, with hands-on experience integrating that software with real hardware and sensors. Delivered a production machine-learning project and a full-stack asset-management platform for PT Kereta Api Indonesia, and completed a research internship at BRIN. Seeking a software engineering role in embedded systems or IoT.

SKILLS

Languages: Python, C, C++, JavaScript, HTML/CSS, MATLAB

Software Development: Computer vision (OpenCV), machine learning (TensorFlow/TFLite Micro), FastAPI, REST API integration, WebSocket, JWT authentication, Firebase Realtime Database, Android Studio, Vercel

Embedded & IoT: ESP32/ESP32-S3, Arduino/PlatformIO, MQTT, HTTP, UART, I2C, sensor and actuator integration (load cells, IMU, gas sensors), PID closed-loop control, LVGL, TFT touchscreen HMI

Tools & Platforms: Git, GitHub, Linux, Visual Studio Code, ROS 2, KiCad, Proteus, MATLAB/Simulink

Spoken Languages: Indonesian (native), English (professional working, EPRT 542), Mandarin (elementary), Japanese (elementary)

EXPERIENCE

Software Engineering Intern (Fullstack)

Jul 2026 - Present

PT KAI (Kereta Api Indonesia) -- Track & Bridge Division — Bandung, Indonesia

Tech: Python, FastAPI, SQL, WebSocket, JWT authentication, JavaScript

- Architecting and building a full-stack asset-management platform (SIMA-KAI) for PT KAI's Track & Bridge division end to end, designed to catalog 103 equipment types across 254 work units in 13 regions nationwide; team of 2, 46+ commits across 21 branches to date.
- Implemented the FastAPI backend's WebSocket live-update layer, JWT authentication and rate limiting, and response-compression middleware cutting a 470 KB JS / 383 KB HTML payload roughly 8:1.

Research Assistant, Programming Division

Feb 2024 - Present

LumiaTech Innovations Research Laboratory, Telkom University — Bandung, Indonesia

Tech: Python, C++, OpenCV, TensorFlow, image segmentation

- Delivered a railway-track anomaly detection system for PT Kereta Api Indonesia, applying OpenCV image segmentation to trackside video to flag defects along the rail corridor; completed and accepted in a team of three.
- Built a multi-view reconstruction module recovering multiple usable viewpoints from a single monocular fisheye camera.

Software Engineering Intern

Jul 2025 - Sep 2025

BRIN (National Research and Innovation Agency), KST Samaun Samadikun — Bandung, Indonesia

Tech: Python, ROS 2, Linux, NVIDIA Isaac Sim

- Built the data collection and processing pipeline for an autonomous vehicle research platform, capturing and synchronising multi-sensor data (LiDAR, IMU, GPS) across multiple recording runs on ROS 2.
- Set up a physics-based simulation environment in NVIDIA Isaac Sim so the team could test control software without on-track vehicle time.

PROJECTS

SkyNES — Custom NES Handheld (ESP32-S3)

Jun 2026 - Present

Tech: C++, ESP32-S3, 6502 CPU emulation, PPU, PlatformIO

- Solo-built a from-scratch NES emulator core — 6502 CPU, PPU, and cartridge/mapper handling — targeting a custom handheld on ESP32-S3 with a 4.0" TFT touchscreen; verified the CPU byte-for-byte against the nestest golden reference trace across all 8,990 instructions.
- Hardware bring-up (display wiring) still in progress; core logic runs correctly end-to-end on the desktop test harness today.
- Repository: github.com/EintsWaveX/SkyNES

Smart Rack Monitoring System

Jan 2026 - Jun 2026

Tech: ESP32, ESP32-S3, LVGL, HX711 load cells, MPU6050, UART, C++

- Personally engineered both firmware halves of a dual-ESP32 industrial rack-monitoring system for a Capstone Design project (5 credits): an ESP32-S3 driving a 480×320 LVGL touchscreen dashboard, and a sensor node reading four HX711 load cells (one per rack footing) plus an MPU6050 IMU over UART.
- Detects up to 200+ kg total load (± 10 -15 kg accuracy) across all four corners, visualized as a live circular gauge plus per-corner readouts, with color, LED, and buzzer overload alerts.
- Repository: github.com/EintsWaveX/SmartRackMonitoring

Community Air Quality Monitoring System — Sindangkerta

2025 - 2026

Tech: ESP32, gas sensors (PM, O₃, NO₂, H₂S, CO), Firebase, Vercel

- Delivered a five-sensor air-quality station (PM2.5/PM10, O₃, H₂S, CO, NO₂) for Sindangkerta village as a completed community-service (pengmas) initiative, streaming live readings to Firebase and a public web dashboard deployed on Vercel.
- Live dashboard: web-pengmas.vercel.app

Intelligent Waste Sorting System

Nov 2025 - Dec 2025

Tech: Python, TensorFlow, TFLite Micro, ESP32-CAM, Firebase, Android Studio

- Built an end-to-end waste-sorting system: a custom lightweight CNN (TensorFlow, INT8-quantized to TFLite Micro) running on an ESP32-CAM classifies plastic, paper, and metal from a live camera feed at 10-150 ms inference latency, averaging over 80% accuracy across four tested input resolutions (up to 91% on plastic, 96% on paper); a servo-driven rotating platform then physically sorts the item into its bin. Owned the system end to end across ML, firmware, and app, in a team of three.
- Synced live device state to Firebase Realtime Database (50-150 ms latency) for a custom Android monitoring app.
- Repositories: [TrainingWorkspace](#) • [IoTxAI](#) • [ES](#)

Two-Chamber PID Temperature Control Prototype

Apr 2025 - Jun 2025

Tech: C++, PID control, DHT22, TFT touchscreen, Arduino

- Designed and tuned a PID closed-loop temperature-control system for a two-chamber aquarium enclosure: a heat gun warms one side (DHT22-sensed) while a 12V DC fan draws chilled water vapor across it, reaching setpoints from room temperature up to 100°C+ with a 2-3 s response, no overshoot, and ± 0.5 -1°C steady-state error.
- Owned PID tuning, sensor/actuator integration, and the 3.2" TFT touch control interface as the primary contributor on a team project (SNA × SKD coursework — Sensors & Actuators × Basic Control Systems).
- Repository: github.com/EintsWaveX/EL4705_09_TugasBesarSNAxSKD

TEACHING & ACADEMIC LEADERSHIP

Practicum Assistant & Division Coordinator

Sep 2024 - Present

Basic Computing Laboratory (C) and IRIS Laboratory (IoT), Telkom University — Bandung, Indonesia

- Author C and IoT programming problems, test cases, and grading rubrics each semester; lead the Assignment and Task Committee, and ran the academic track of DLOR 2025 assistant recruitment.

Lecturer's Assistant — Digital Engineering, Calculus I

May 2025 - Jan 2026

Telkom University — Bandung, Indonesia

- Taught and assessed two undergraduate courses, designing quizzes, examinations, and projects.

EDUCATION

Bachelor of Engineering, Electrical Engineering

Expected Sep 2027

Telkom University — Bandung, Indonesia

- GPA 3.77/4.00. Relevant coursework: Digital and Microcomputer Engineering, Artificial Intelligence and Machine Learning, Embedded Systems, Control Systems, Internet of Things.